MTH320 2021S2 Exam

Feel free to contact me at maths_and_programming@outlook.sg or @geodome on Telegram if you need tuition.

I provide undergraduate level tuition for the following SUSS units:

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- MTH210 Fundamentals of Probability's
- MTH212 Statistical Analysis
- MTH240 Engineering Mathematics 1
- MTH316 Multivariable Calculus
- MTH321 Engineering Mathematics 2
- MTH355 Basic Mathematical Optimisation

Donaldson Tan

MTH320

Timed Online Assignment – July Semester 2021

Mathematical Methods II

Friday, 26 November 2021 **4:00 pm – 6:30 pm**

Time allowed: 2.5 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **FIVE (5)** questions and comprises **FIVE (5)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, your Full Name, and SUSS PI Number. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT Space. *Example: TOAXYZ_RaphaelLee_T1923161 (omit D/O, S/O).*

6. Your submission should consist of only one file and must not exceed 500MB in size. The file must be a PDF or Microsoft Word file saved in .doc or .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the PDF or Word file provided it does not exceed the file size limit of 500MB.

7. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. All uploaded hand-written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

8. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

9. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

10. Marks will only be awarded if FULL working is shown.

Answer all questions. (Total 100 marks)

Question 1

A system of 2 ODEs with initial conditions is given as follows.

$$y_1' = 6y_1 - y_2, \quad y_1(0) = 1$$
$$y_2' = 2y_1 + 3y_2 + 1, \quad y_2(0) = 0$$

(a) Rewrite the system of ODEs and their initial conditions in matrix notation.

(5 marks)

(b) Using the Duhamel's principle, solve the system of ODEs.

(15 marks)

Question 2

Figure Q2 shows a torpedo of mass, m, travelling horizontally underwater with a propulsion force P(t). It experiences a drag force which is proportional to its velocity with a constant of proportionality, c.

<u>Figure Q2: Torpedo travelling with position function, x(t)</u>

(a) Sketch a free-body diagram of the torpedo with all forces clearly labelled. By applying Newton's 2nd law on the torpedo, show that the governing equation for the horizontal motion of the torpedo is:

$$x'' + \frac{c}{m}x' = \frac{P(t)}{m}.$$

(5 marks)

(b) After the launch of the torpedo, the propulsion force has the expression:

$$P(t) = Q\left(1 - e^{-t}\right)$$

where Q is a constant. Using the method of undetermined coefficients, solve the governing equation in Question 2(a) and show that the general solution for the position of the torpedo is:

$$x(t) = C_1 + C_2 e^{-\frac{c}{m}t} + \frac{Q}{c}t + \frac{Q}{c-m}e^{-t}$$

where C_1 and C_2 are constants. Note that $c \neq m$.

(15 marks)

(c) Hence, state the steady-state motion of the torpedo and explain.

(7 marks)

Question 3

A damped mass-spring system has the following equation of motion. Determine the particular solution using Laplace transform. Is the system underdamped, critically damped or overdamped ?

$$y'' + 6y' + 10y = \delta(t-3),\ y(0) = 0,\ y'(0) = 1$$

(15 marks)

Question 4

(a) Compute the Fourier series representation of:

$$f(x) = -5x,\ \ -2 < x < 2.$$

(9 marks)

(b) Hence or otherwise, solve the heat conduction problem along a 2-meter rod described by:

$$\begin{aligned} \alpha T_{xx} &= T_t \qquad\qquad , 0 < x < 2,\ t > 0 \\ T(0,t) = 30,\ \ T(2,t) &= 18,\ t > 0 \\ T(x,0) &= 30 - 11x \qquad , 0 < x < 2 \end{aligned}$$

(14 marks)

Question 5

Solve the wave equation problem on a 2-meter long string described by:

$$\begin{aligned} c^2 u_{xx} &= u_{tt} && , \ 0 < x < 2, \ t \geq 0 \\ u(0,t) = 0, \ \ u(2,t) &= 0 && , \ t \geq 0 \\ u(x,0) = 1 - (x-1)^2, \ u_t(x,0) &= 1, \ 0 < x < 2 \end{aligned}$$

(15 marks)

----- END OF PAPER -----

1(a) $\begin{bmatrix} y_1' \\ y_2' \end{bmatrix} = \begin{bmatrix} 6 & -1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$\begin{bmatrix} y_1(0) \\ y_2(0) \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

(b) $y' = \begin{bmatrix} y_1' \\ y_2' \end{bmatrix}$ $A = \begin{bmatrix} 6 & -1 \\ 2 & 3 \end{bmatrix}$ $f(t) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$y(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

let $\det(A - \lambda I) = 0$

$\begin{vmatrix} 6-\lambda & -1 \\ 2 & 3-\lambda \end{vmatrix} = 0$

$(6-\lambda)(3-\lambda) - (-1)(2) = 0$

$\lambda^2 - 9\lambda + 20 = 0$

$(\lambda - 5)(\lambda - 4) = 0$

$\lambda = 5, 4$

case $\lambda = 4$: $\left[\begin{array}{cc|c} 6-\lambda & -1 & 0 \\ 2 & 3-\lambda & 0 \end{array}\right] \xrightarrow{\lambda=4} \left[\begin{array}{cc|c} 2 & -1 & 0 \\ 2 & -1 & 0 \end{array}\right] \longrightarrow \left[\begin{array}{cc|c} 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 \end{array}\right]$

$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{2}x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix}$

eigenvector $v_1 = 2\begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

case $\lambda = 5$: $\left[\begin{array}{cc|c} 6-\lambda & -1 & 0 \\ 2 & 3-\lambda & 0 \end{array}\right] \xrightarrow{\lambda=5} \left[\begin{array}{cc|c} 1 & -1 & 0 \\ 2 & -2 & 0 \end{array}\right] \longrightarrow \left[\begin{array}{cc|c} 1 & -1 & 0 \\ 0 & 0 & 0 \end{array}\right]$

$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

eigenvector $v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\therefore A = PDP^{-1}$ where $P = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$, $D = \begin{bmatrix} 4 & 0 \\ 0 & 5 \end{bmatrix}$, $P^{-1} = \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix}$

$e^{At} = Pe^{Dt}P^{-1}$

$= \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} e^{4t} & 0 \\ 0 & e^{5t} \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix}$

$= \begin{bmatrix} 2e^{5t} - e^{4t} & -e^{5t} + e^{4t} \\ 2e^{5t} - 2e^{4t} & -e^{5t} + 2e^{4t} \end{bmatrix}$

1(b) $e^{A(t-t_0)}\vec{y}(t_0) = e^{At}\vec{y}(0)$

$$= \begin{bmatrix} 2e^{5t} - e^{4t} \\ 2e^{5t} - 2e^{4t} \end{bmatrix}$$

$$e^{A(t-s)}\vec{f}(s) = \begin{bmatrix} 2e^{-5s+5t} - e^{-4s+4t} & -e^{-5s+5t} + e^{-4s+4t} \\ 2e^{-5s+5t} - 2e^{-4s+4t} & -e^{-5s+5t} + 2e^{-4s+4t} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 2e^{-5s+5t} - e^{-4s+4t} \\ 2e^{-5s+5t} - 2e^{-4s+4t} \end{bmatrix}$$

$$\int_0^t e^{A(t-s)}\vec{f}(s)\,ds = \int_0^t \begin{bmatrix} 2e^{-5s+5t} - e^{-4s+4t} \\ 2e^{-5s+5t} - 2e^{-4s+4t} \end{bmatrix} ds$$

$$= \left.\begin{bmatrix} -\frac{2}{5}e^{-5s+5t} + \frac{1}{4}e^{-4s+4t} \\ -\frac{2}{5}e^{-5s+5t} - \frac{1}{2}e^{-4s+4t} \end{bmatrix}\right|_{s=0}^{s=t}$$

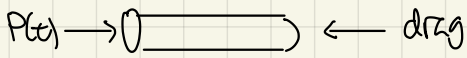
$$= \begin{bmatrix} \frac{2}{5}e^{5t} - \frac{1}{4}e^{4t} - \frac{3}{20} \\ \frac{2}{5}e^{5t} - \frac{1}{2}e^{4t} + \frac{1}{10} \end{bmatrix}$$

By Duhammel's Principle,

$$\begin{bmatrix} y_1(t) \\ y_2(t) \end{bmatrix} = e^{At}\vec{y}(0) + \int_0^t e^{A(t-s)}\vec{f}(s)\,ds$$

$$= \begin{bmatrix} \frac{12}{5}e^{5t} - \frac{5}{4}e^{4t} - \frac{3}{20} \\ \frac{12}{5}e^{5t} - \frac{5}{2}e^{4t} + \frac{1}{10} \end{bmatrix}$$

2(a)



$$\sum F = P(t) - c\,x'$$

$$m\,x'' = P(t) - c\,x'$$

$$P(t) = m\,x'' + c\,x'$$

$$\frac{P(t)}{m} = x'' + \frac{c}{m}x'$$

(b) $P(t) = Q(1-e^{-t})$

$$\frac{Q}{m}(1-e^{-t}) = x'' + \frac{c}{m}x'$$

$$\frac{Q}{m} - \frac{Q}{m}e^{-t} = x'' + \frac{c}{m}x'$$

let $x''_{CF} + \frac{c}{m}x'_{CF} = 0$

$$s^2 + \frac{c}{m}s = 0$$

$$s\left(s + \frac{c}{m}\right) = 0$$

$$s = 0 \text{ or } -\frac{c}{m}$$

$\therefore x_{CF} = C_1 e^{0t} + C_2 e^{-\frac{c}{m}t} = C_1 + C_2 e^{-\frac{c}{m}t}$ where C_1, C_2 are constants

$$x(t) = C_1 + C_2 e^{-\frac{c}{m}t} + \frac{Q}{c}t + \frac{Q}{c-m}e^{-t}$$

general solution given in question

infer $x_{PI} = D_1 t + D_2 e^{-t}$

let $x_{PI} = D_1 t + D_2 e^{-t}$

$$x'_{PI} = D_1 - D_2 e^{-t}$$

$$x''_{PI} = D_2 e^{-t}$$

$$x''_{PI} + \frac{c}{m}x'_{PI} = D_2 e^{-t} + \frac{c}{m}(D_1 - D_2 e^{-t})$$

$$= \frac{c}{m}D_1 + D_2\left(1 - \frac{c}{m}\right)e^{-t}$$

$$\Rightarrow \frac{c}{m}D_1 = \frac{Q}{m} \Rightarrow cD_1 = Q \Rightarrow D_1 = \frac{Q}{c}$$

$$\Rightarrow D_2\left(1 - \frac{c}{m}\right) = -\frac{Q}{m}$$

$$D_2 = -\frac{Q}{m} \div \left(1 - \frac{c}{m}\right)$$

$$= -\frac{Q}{m} \div \frac{m-c}{m}$$

$$= \frac{Q}{c-m}$$

$$\therefore x(t) = x_{CF}(t) + x_{PI}(t)$$

$$= C_1 + C_2 e^{-\frac{c}{m}t} + D_1 t + D_2 e^{-t}$$

$$= C_1 + C_2 e^{-\frac{c}{m}t} + \frac{Q}{c}t + \frac{Q}{c-m}e^{-t}$$

(c) As t approaches infinity, all the exponential terms in $x(t)$ become 0

Hence, steady state motion $x_{steady}(t) = C_1 + \frac{Q}{c}t$

3 $y'' + 6y' + 10y = \delta(t-3),\ y(0) = 0,\ y'(0) = 1$

$$\mathcal{L}\{y'' + 6y' + 10y\} = \mathcal{L}\{\delta(t-3)\}$$

$$(s^2Y(s) - sy(0) - y'(0)) + 6(sY(s) - y(0)) + 10Y(s) = e^{-3s}$$

$$(s^2Y(s) - 1) + 6sY(s) + 10Y(s) = e^{-3s}$$

$$(s^2 + 6s + 10)Y(s) = 1 + e^{-3s}$$

$$Y(s) = \frac{1+e^{-3s}}{s^2+6s+10}$$

$$= \frac{1+e^{-3s}}{(s+3)^2+1^2}$$

$$= \frac{1}{(s+3)^2+1} + \frac{e^{-3(s+3)}e^{9}}{(s+3)^2+1}$$

$$= \left[\frac{1}{s^2+1} + \frac{e^{9}e^{-3s}}{s^2+1}\right]_{s \to s+3}$$

$$y(t) = e^{-3t}\mathcal{L}^{-1}\left\{\frac{1}{s^2+1} + \frac{e^{9}e^{-3s}}{s^2+1}\right\}$$

$$= e^{-3t}\mathcal{L}^{-1}\left\{\frac{1}{s^2+1}\right\} + e^{9-3t}\mathcal{L}^{-1}\left\{\frac{e^{-3s}}{s^2+1}\right\}$$

$$= e^{-3t}\sin(t) + e^{9-3t}u(t-3)f(t-3) \quad \text{where } f(t) = \mathcal{L}^{-1}\left\{\frac{1}{s^2+1}\right\} = \sin(t)$$

$$= e^{-3t}\sin(t) + e^{9-3t}u(t-3)\sin(t-3)$$

4(a) $f(x) = -5x$ for $-2 \le x \le 2$

$f(x+4) = f(x)$

$T = 2-(-2) = 4$

$\omega = 2\pi / T = \frac{\pi}{2}$

$-f(-x) = -5(-x) = -5x = f(x)$

$\Rightarrow f(x)$ is an odd function

$\Rightarrow a_0 = a_n = 0$

$$b_n = \frac{2}{4}\int_{-2}^{2} f(x)\sin(n\omega x)\,dx$$

$u = x \quad du = 1$

$dv = -\sin\frac{n\pi x}{2} \quad v = \frac{2}{n\pi}\cos\frac{n\pi x}{2}$

$$= \frac{1}{2}\int_{-2}^{2} -5x\sin\left(\frac{n\pi x}{2}\right)dx$$

$$= \frac{5}{2}\int_{-2}^{2} x \cdot -\sin\left(\frac{n\pi x}{2}\right)dx$$

$$= \frac{5}{2}\left[\frac{2x}{n\pi}\cos\left(\frac{n\pi x}{2}\right) - \int \frac{2}{n\pi}\cos\left(\frac{n\pi x}{2}\right)dx\right]_{-2}^{2}$$

$$= \frac{5}{2}\left[\frac{2x}{n\pi}\cos\left(\frac{n\pi x}{2}\right) - \left(\frac{2}{n\pi}\right)^2\sin\left(\frac{n\pi x}{2}\right)\right]_{-2}^{2}$$

$$= \frac{5}{2}\left[\frac{4}{n\pi}\cos(n\pi) - \left(-\frac{4}{n\pi}\cos(n\pi x)\right)\right]$$

$$= \frac{5}{2}\cdot\frac{8}{n\pi}\cos(n\pi)$$

$$= \frac{20}{n\pi}(-1)^n$$

$$\therefore f(x) = \sum_{n=1}^{\infty} b_n \sin(n\pi x)$$

$$= \sum_{n=1}^{\infty} \frac{20}{n\pi}(-1)^n \sin(n\pi x)$$

4(b)

$$\alpha T_{xx} = T_t \quad , 0 < x < 2,\ t > 0$$
$$T(0,t) = 30,\ \ T(2,t) = 18,\ t > 0$$
$$T(x,0) = 30 - 11x \quad , 0 < x < 2$$

Heat Equation with Fixed End Temperature

$$T_s(x) = 30 - \frac{30-18}{2}x = 30 - 6x \quad , \ L = 2$$

To find transient state solution

$$\alpha Q_{xx} = Q_t$$
$$Q(0,t) = T(0,t) - T_s(0) = 0$$
$$Q(2,t) = T(2,t) - T_s(2) = 0$$
$$Q(x,0) = T(x,0) - T_s(x) = (30 - 11x) - (30 - 6x) = -5x$$

$$\Rightarrow Q(x,t) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right) e^{-\alpha\left(\frac{n\pi}{L}\right)^2 t}$$
$$= \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{2}\right) e^{-\alpha\left(\frac{n\pi}{2}\right)^2 t}$$
$$= \sum_{n=1}^{\infty} \frac{20}{n\pi}(-1)^n \sin\left(\frac{n\pi x}{2}\right) e^{-\alpha\left(\frac{n\pi}{2}\right)^2 t}$$

$$\therefore T(x,t) = Q(x,t) + T_s(x)$$
$$= \sum_{n=1}^{\infty} \frac{20}{n\pi}(-1)^n \sin\left(\frac{n\pi x}{2}\right) e^{-\alpha\left(\frac{n\pi}{2}\right)^2 t} + 30 - 6x$$

5

$$c^2 u_{xx} = u_{tt} \qquad , \ 0 < x < 2, \ t \geq 0$$
$$u(0,t) = 0, \ \ u(2,t) = 0 \qquad , \ t \geq 0$$
$$u(x,0) = 1 - (x-1)^2, \ u_t(x,0) = 1, \ 0 < x < 2$$

(15 marks)

$L = 2$

$$a_n = \frac{2}{L}\int_0^L u(x,0)\sin\left(\frac{n\pi x}{L}\right)dx$$

$$= \frac{2}{2}\int_0^2 \left[1-(x-1)^2\right]\sin\left(\frac{n\pi x}{2}\right)dx$$

$$= \int_0^2 \underbrace{(2x-x^2)}_{u}\underbrace{\sin\left(\frac{n\pi x}{2}\right)}_{dv}dx$$

$du = 2-2x \qquad v = -\frac{2}{n\pi}\cos\left(\frac{n\pi x}{2}\right)$

$$= \left[(2x-x^2)\left(-\frac{2}{n\pi}\right)\cos\left(\frac{n\pi x}{2}\right)\right]_0^2 - \int_0^2 -\frac{2}{n\pi}\cos\left(\frac{n\pi x}{L}\right)(2-2x)\,dx$$

$$= 0 + \frac{4}{n\pi}\int_0^2 \underbrace{(1-x)}_{u}\cos\left(\frac{n\pi x}{2}\right)dx$$

$du = -1 \qquad dv = \cos\left(\frac{n\pi x}{2}\right) \qquad v = \frac{2}{n\pi}\sin\left(\frac{n\pi x}{2}\right)$

$$= \frac{4}{n\pi}\left[\frac{2(1-x)}{n\pi}\sin\left(\frac{n\pi x}{2}\right) + \int \frac{2}{n\pi}\sin\left(\frac{n\pi x}{2}\right)dx\right]_0^2$$

$$= \frac{4}{n\pi}\left[\frac{2(1-x)}{n\pi}\sin\left(\frac{n\pi x}{2}\right) + \left(\frac{2}{n\pi}\right)^2\int \frac{n\pi}{2}\sin\left(\frac{n\pi x}{2}\right)dx\right]_0^2$$

$$= \frac{4}{n\pi}\left[\frac{2(1-x)}{n\pi}\sin\left(\frac{n\pi x}{2}\right) - \left(\frac{2}{n\pi}\right)^2\cos\left(\frac{n\pi x}{2}\right)\right]_0^2$$

$$= \frac{4}{n\pi}\left[\left(0 - \left(\frac{2}{n\pi}\right)^2\cos(n\pi)\right) - \left(0 - \left(\frac{2}{n\pi}\right)^2\right)\right]$$

$$= \frac{4}{n\pi}\cdot\left(\frac{2}{n\pi}\right)^2\cdot\left(1-\cos(n\pi)\right)$$

$$= \frac{16}{n^3\pi^3}\left(1-(-1)^n\right)$$

5. $$b_n = \frac{2}{n\pi c}\int_0^2 u_t(x,0)\sin\left(\frac{n\pi x}{2}\right)dx$$

$$= \frac{2}{n\pi c}\int_0^2 \sin\left(\frac{n\pi x}{2}\right)dx$$

$$= -\frac{2}{n\pi c}\cdot\frac{2}{n\pi}\int_0^2 -\frac{n\pi}{2}\sin\left(\frac{n\pi x}{2}\right)dx$$

$$= -\frac{4}{n^2\pi^2 c}\left[\cos\left(\frac{n\pi x}{2}\right)\right]_0^2$$

$$= -\frac{4}{n^2\pi^2 c}\left[\cos(n\pi) - 1\right]$$

$$= \frac{4}{n^2\pi^2 c}\left[1-(-1)^n\right]$$

$$\therefore u(x,t) = \sum_{n=1}^{\infty}\sin\left(\frac{n\pi x}{2}\right)\left[\frac{16}{n^3\pi^3}(1-(-1)^n)\cos\left(\frac{n\pi ct}{2}\right) + \frac{4}{n^2\pi^2 c}\left(1-(-1)^n\right)\right]$$